

Towards On-the-wall Tangible Interaction: Using Walls as Interactive, Dynamic, and Responsive User Interface

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ABSTRACT

This paper presents our vision of *on-the-wall tangible interaction*. We envision a future where tangible interaction can be extended from conventional horizontal surfaces to vertical surfaces; indoor vertical areas such as walls, windows, and ceilings can be used for dynamic and direct physical manipulation. We first discuss the unique properties that vertical surfaces may offer for tangible interaction and the interaction scenarios they imbue. We then propose two potential paths for realizing on-the-wall interaction and the technical challenges we face. We follow with one prototype called Climbot. We showcase how Climbot can be used as an on-the-wall tangible user interface for dynamic lighting and as a wall switch controller. We conclude with a discussion of future work.

CCS CONCEPTS

• **Human-centered computing**; • **Human Computer Interaction (HCI)**; • **Interaction Diagram**; • **Graphical user interfaces**; • **Ubiquitous and mobile computing**; • **Ubiquitous and mobile computing theory, concepts and paradigms**; • **Ubiquitous computing**;

KEYWORDS

Tangible User Interface, Tangible Interaction, Wall Interaction, Wall, Robotics, Climbing Robot

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1 VISION

Tangible User Interface (TUI) [7, 9] has been expanding with advancements in AI, robotics, material science, etc. Researchers have explored tangible interactions on flat surfaces small [13, 22] and large [1, 21], with 2.5D displays that can change the heights [4, 24], and with actuated tangibles with relocation capabilities [12, 16, 18]. While extensively developed, most TUI systems are limited to surfaces that are horizontal, such as tabletops or floors [1, 10, 28]. Walls, substrates that are ubiquitous in almost all architectures and our lives, are rarely used for interactive purposes, except when equipped with pixel-based 2D displays [6, 19]. We see this as an overlooked opportunity. In this work, we propose to extend tangible interactions from horizontal surfaces to vertical surfaces. We envision a future where fixtures on vertical surfaces, such as walls and windows, will be dynamic, responsive, and interactive. They may change their shapes, locations, and behaviors according to the user's needs and environmental input. We call it *on-the-wall tangible interaction*. As the first step towards this vision, in this paper, we discuss the uniqueness of walls as substrates for tangible interaction, the potential ways to design and build on-the-wall TUIs, and demonstrate one exemplar implementation – Climbot, which achieves dynamic on-the-wall lighting and switch control.

1.1 Beyond Just Walls

Walls are ubiquitous in modern lives. They serve as the foundations to hold up construction and as placeholders for various household essentials such as lightings, mirrors, switches, and TV displays. Although walls have not been used much for tangible interactions, we see the great potential that can make walls beyond passive building infrastructures. Here we consider four properties of walls that can transform these vertical substrates to be active, interactive surfaces.

- *Walls offer a large interaction area.* In most indoor residential and office rooms, walls and windows take up more than half of the indoor space and usually act as passive structural constructs. Their functions are mostly restricted to hide infrastructure, wires, and pipes or to provide an optical view to the outdoors, making these structures' surfaces largely ornamental. These large, flat, and unused indoor areas will be resourceful if any TUI system can utilize them for interactions.

- *Walls can be a unique channel for interactions with minimal interference to other functions of the space.* Tangible interfaces will inevitably include physical actuators or moving objects. These physical objects may interfere with the operations of the substrates in a lot of ways. For example, in an indoor environment, the floor can be challenging to build a proper TUI considering the spatial conflict of people walking. The relocation or morphing of the tangibles must be carefully designed to avoid competing with the user sharing the same space [25]. Walls and vertical surfaces, on the other hand, have enough space so as to not interfere with everyday human activities. Thus, wall surfaces can provide a stand-alone channel for tangible interactions.
- *Wall fixtures provide rich interaction opportunities.* Recent wall fixtures such as lightings, multimedia systems, switches, and cabinets have been equipped with IoT features that enable sensing and remote controls [11, 15, 32]. However, they are all permanently fixed to predetermined positions on the wall. We consider an enhanced version of future wall artifacts, where they are not only 'smart' IoT devices but can also relocate themselves on the walls. For example, a lighting system or a multimedia sound system can be adaptable to where the users are in order to create an optimized lighting layout or surrounding sound effects; wall outlets and switch panels are on-demand so that they can approach users instead of having users walk up to them; cabinets can be dynamic and locate themselves to higher-up ceilings when not required and move to the desired location when summoned by user. With mobility in the picture, existing fixtures on walls will provide next-level indoor experiences.
- *Existing tangible interaction technologies can make use of walls to create an experience on another level.* Many existing technologies developed for other surfaces like tabletops or screens can also elevate the interactive experience when adapted to walls. For example, TRANSFORM [8] is a tabletop 2.5D display built upon pin-based actuators. With the technology applied to walls, a shape-changing wall surface across the room can support more interactions, such as dynamically generated seating areas or hangers extended from the wall. Venous Materials [14], as a microfluid structure, is also adaptable to walls. The distant hydraulic transmission of fluid is extensible for remote information delivery between one spot on the wall to a different area higher up on the ceiling or even another room. Other works [5, 17] also make great examples for bringing existing interaction techniques like capacitive [31] and acoustic sensing into large vertical areas like walls and windows.

1.2 Ultimate Tangible Interaction on the Wall

To better illustrate our vision of on-the-wall tangible interaction, we present an exemplar scenario of a day in the life of Taylor, a future knowledge worker. We demonstrate how vertical tangible interfaces can be blended into Taylor's life with both residential and workspace contexts. We also showcase scenarios where TUI on the wall can be beneficial for health care and education.

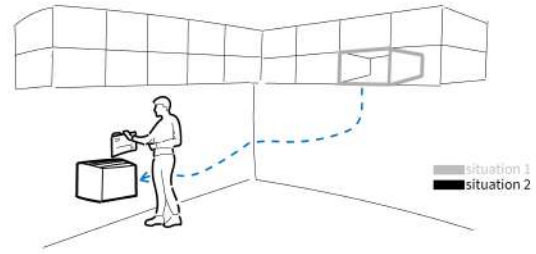


Figure 1: Mobile storage cabinets.

Taylor wakes up in the morning and issues command for today's clothes via smart home-keeper. The cloth cabinet, equipped with on-the-wall robots as active tangibles, lowers from the ceilings and brings the outfits alongside the wall. After breakfast, a coat hanger comes right next to the dressing mirror with coats and hats. The mirror is a camera-based display hidden in the wall. It only acts as a mirror when Taylor presents in front of the area.

When Taylor arrives at the office, a pin-based control panel emerged from the office wall. With one click, Taylor has the table extended out of the wall, and the area of the wall becomes a display. This display acts as a 3D tangible display when needed, making the meeting with remote colleague Alex easy and smooth. As a designer, Alex presents the design with a pin-based manipulator synced with Taylor's on-the-wall 3D display that will shape the same 3D form factor as Alex pursues. Part of Taylor's job is to document the product's designs and organize the prototypes in a warehouse. Documents are stored in a dynamic cabinet system in the office (Figure 1). The required cabinet will be transferred to Taylor when needed (again with on-the-wall robots), and after manipulation, go back to the ceiling, leaving the lower-level space free for other activities. The warehouse also utilizes similar tangible systems. When Taylor arrives, the warehouse will allocate a robot on the racks to navigate Taylor to the designated location.

When Taylor is finally home, cloth hangers will come to the gate on the wall for the coat and hat after a day of work. The light above the entrance will light up and follow the steps into the rooms along with the ceiling. The soundtrack played is multidirectional (Figure 2), with on-the-wall robots carrying the speakers around, adapting to Taylor's real-time location and posture.

Taylor's family also benefits from on-the-wall tangible interactions. Taylor's partner works in a hospital; For emergency cases, medical and surgical supplies are delivered to the emergency room utilizing the on-the-wall tangible system. These tangibles use walls as the travel substrate to avoid the crowds in the hospital hallways and can move at high speed, ensuring emergency supplies arrive in time. Taylor's child is in high school; students enjoy a multimedia learning environment with the multipurpose blackboard (Figure 3). The lecturer can still use the blackboard for writing, while any part of the blackboard can be used as a high-resolution 3D display to present topic-relevant contents.

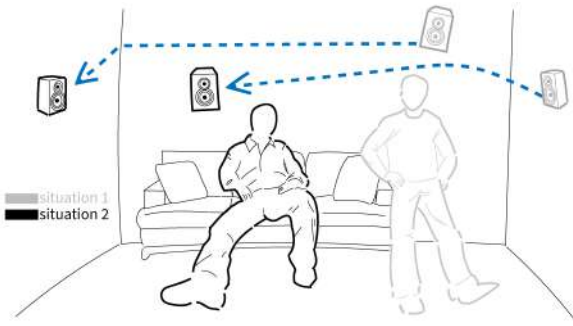


Figure 2: Dynamic sound system enabled by on-the-wall mobile speakers.

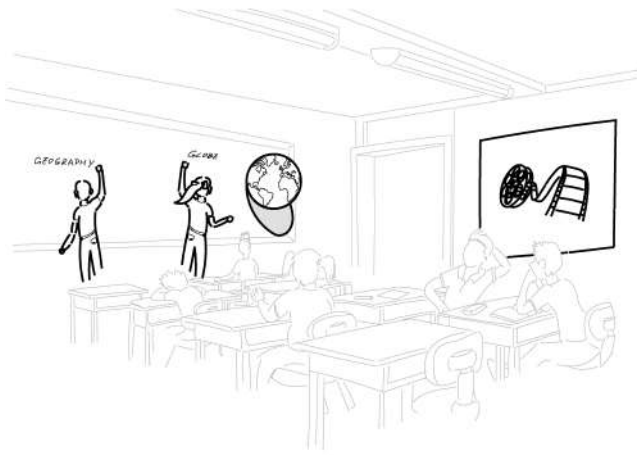


Figure 3: Multipurpose blackboard as a 3D display.

2 CHALLENGES FOR REALIZATION

As illustrated above, we mainly consider two potential approaches for on-the-wall tangible interactions: 1) make the walls substrate smart with shape-changing capabilities, and 2) use on-the-wall robots to introduce mobility to existing wall fixtures. Here we will discuss the technical challenges for realizing the two types of TUIs on the wall.

2.1 Making Wall Substrate Dynamic

There are several challenges to build walls with substrates that can change shape. First, construction cannot compromise the structural demand of the walls. Certain modifications to the wall surface or its frame may cause the weakening of the walls' load capacity, which must not be sacrificed or may cause safety concerns to the indoor environment. Second, to create vertical interfaces with dynamic behavior, the underlying actuators need to bear more downward loads. For example, linear actuators on the wall being used as hooks or furniture such as chairs will sustain much more force than those used on tabletop interfaces as the extended area will act as a lever. Third, the hardware setup has to be space conservative. According

to the widely used stud frame or brick-based wall structures, embeddings can only happen within a thickness of tens of centimeters [26, 31]. Space thus may not be sufficient for many existing actuator mechanisms, such as pneumatic actuators requiring a pump or hydraulic linear actuators requiring a matrix of control devices and tubes. Finally, the design and engineering effort of integrating all sensors, actuators, and existing wall structures cannot be negligent.

2.2 Making Wall Fixtures Dynamic

Using on-the-wall robots can make fixtures mobile. However, on-the-wall robots face a different set of challenges, mainly countering the gravity effect. On-the-wall artifacts include small objects like switches and controls as well as heavy lifting items like lights, TVs, speakers, etc. The carrier robots that hold and move these fixtures across walls and ceilings thus have to fight against a considerable downforce. Multiple solutions have been explored, while each has its limitation. Several work [2, 27] explores a wall-based robot using a cable-driven mechanism for vertical surface locomotion. While the robot can bear a heavy load, the moving area is greatly limited by the tendon cable setup. Micro and nano-scaled structures [30] would also provide good adhesion for on-the-wall robots, while the substrate surface roughness is limited in realization. Suction-based climbing robot [23], although having a good payload, usually moves slowly and makes loud noise due to the required pump system. Similarly, propellers' negative pressure also provides considerable stability [20] for a climbing mechanism, but the high energy consumption and loud noise make it unsuitable for indoor usage.

3 CLIMBOT

As the first step towards the ultimate on-the-wall tangible interaction, we present an on-going work that uses an on-the-wall robot to make wall fixtures dynamic. We propose a mobile robot prototype – Climbot, to accomplish dynamic on-the-wall tangible interactions confronting the challenges described above.

3.1 Design Considerations

To achieve the robot-based on-the-wall tangible interaction system, we consider the following design criteria.

- *Stable climbing, turning, and deployment.* The first design consideration is reliable locomotion capability on the walls. To support indoor sensing and interaction, the locomotion capability needs to allow a robot to move along the vertical surface in all directions and be able to turn at different angles. The robot should also be able to remain statically on the wall when no interaction is performed.
- *High payload-body ratio.* Compared to robots used in tabletop or floor-based TUIs, the robot for on-the-wall interaction needs to keep its body lightweight while ensuring the capacity to carry other payloads such as external sensors, actuators, and existing wall fixtures like lighting.
- *Low noise.* Unlike previous wall-climbing mechanisms primarily aimed at hazard rescuing, we focus on indoor on-the-wall interaction. Thus, the robot needs to maintain relatively low noise during the operation without interfering with regular indoor activities.

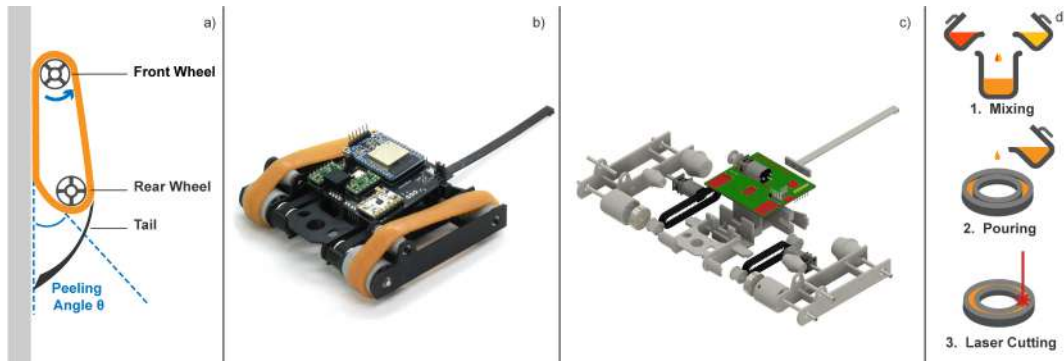


Figure 4: Climbot. a) Illustration of the climbing mechanism; b) Climbot assembly; c) The explosion view of the Climbot design; d) the fabrication process of a tread.

- **Size.** We require our robot to 'blend into' the indoor environment for intuitive interaction scenarios. Thus, the size of the robots has to be considerably small and ideally unnoticeable.

3.2 System Overview

The Climbot is an on-the-wall tangible interaction prototype to introduce mobility to sensors, actuators, and existing on-the-wall fixtures. Following the design criteria mentioned above, we decide to use dry adhesive treads as the primary climbing mechanism. Specifically, our robot design is inspired by Tankbot [29], which utilizes sticky treads to travel freely on the walls and can make far less noise compare to vacuum-based robots [18].

3.3 Climbot Design

Figure 4 demonstrates the explosion view of the Climbot design with its main components. For locomotion, the Climbot uses custom dry adhesive treads to climb and steer on the walls (Figure 4a). The Climbot features a spring-loaded tail that can balance the center mass of the robot to prevent falling. The robot is controlled via WIFI through an onboard ESP32 chip; a webcam a few feet away from the wall is used to sense the robot's motion and location.

The robot body is 3D printed and assembled with commonly used hardware (Figure 4b and c). The dry adhesive tread is fabricated by laser cutting a molded composite rubber material (Vytaflex 10)

(Figure 4d). Fixtures such as light or speakers can be integrated with the Climbot body with easy snap fits.

Climbot climbs on glass, smooth walls, and painted walls. It can reach a speed of 50mm/s when traveling in a straight line. The turning of the robot is enabled by differentiating the speed of the treads. The current prototype's minimum turning radius is 400mm. The stickiness is reinforced by the momentum of the front wheels provided by the motors during locomotion, while the stability of Climbot idling on the wall after 5 minutes is compromised.

3.4 Application Examples

To demonstrate the function and extendibility of the Climbot, we present two simple applications. Figure 5a shows a light carried by the Climbot that is dynamic and can follow the user indoors. The example can be further extended by changing the light to other fixtures such as speakers or displays to achieve a multi-functional responsive indoor environment. Figure 5b shows another example where the robot goes to a switch to turn a lamp off on-demand. Collaborative jobs can potentially be done by synchronizing multiple bots.

3.5 Future Work

We demonstrated a climbing robot for simple on-the-wall tangible interactions. There are several features we are considering for future development.

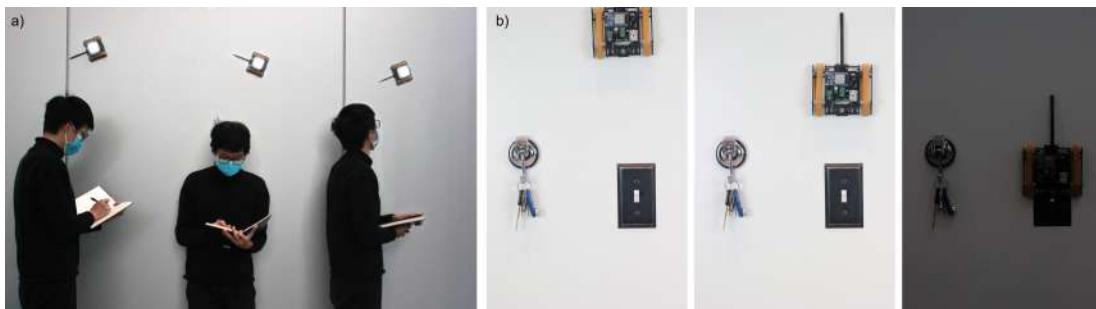


Figure 5: Applications. a) Climbot provides dynamic task lighting following the user. b) Climbot turns off the light switch by walking past the switch control.

- *In-situ Steering*. The current prototype leverages the speed difference between the two treads for steering. Though capable of steering on-the-wall, the mechanism requires a large turning radius, limiting the Climbot's locomotion capability. We will research alternative turning methods to enable in-situ steering with more flexibility.
- *Long-term deployment*. The robot is fighting gravity mainly with the sticky treads. It can climb on the wall reliably with the motor torque but is not stable when idle on the wall. In the future, we will look into mechanisms that can deploy the robot on the wall without active climbing.
- *Multi-Agent Application*. In this work, we use one robot for on-the-wall tangible interaction. In the future, we will explore multiple or swarm robots [3, 12] to expand on-the-wall interactions. For example, multi-agents can be used to create a multidimensional soundtrack with beam steering, a dynamic display with synced positioning, and act as an indoor tracking system for factories or warehouses.

4 CONCLUSION

In this paper, we presented our vision for on-the-wall tangible interaction. We showed the potential for walls being a ubiquitous TUI substrate. We introduced challenges and high-level solutions for developing on-the-wall interfaces. We presented Climbot as a working example of dynamic on-the-wall interaction. Our future work includes refining the concepts around on-the-wall tangible interaction and future Climbot development.

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